

YILAN CITY, Taiwan

WMO: 467080

Lat: 24.7639N

Lon: 121.7565E

Elev: 7

StdP: 101.24

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	11.3	6.3	5.9	11.6	8.0	6.7	12.6	7.8	16.7	7.0	16.9	1.8	250	0.264

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	33.4	28.1	32.9	27.9	32.1	27.6	29.1	32.3	28.5	31.8	28.0	31.3	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.4	31.6	27.8	23.8	31.3	27.1	22.8	30.7	95.3	32.6	92.4	31.8	90.2	31.4	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.2	5.1	8.7	35.9	1.2	1.6	7.8	37.0	7.1	38.0	6.4	38.9	5.5	40.1		
(5)			6.8	30.9	1.6	1.1	5.6	31.7	4.6	32.3	3.7	32.9	2.5	33.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	16.4	16.8	19.2	22.0	24.5	27.1	28.9	28.7	26.8	23.6	21.1	18.3
	DBStd	4.95	2.62	3.07	3.06	3.16	2.31	1.99	0.99	1.07	1.80	2.20	2.57	2.65
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	208	71	62	28	8	1	0	0	0	0	0	6	33
	CDD10.0	4681	198	189	284	360	450	514	587	581	505	422	334	257
	CDD18.3	1848	11	18	54	118	192	264	328	322	255	164	89	33
	CDH23.3	15752	12	36	160	578	1224	2580	3924	3747	2358	801	279	51
	CDH26.7	5268	1	1	13	102	280	863	1638	1481	709	141	37	3
Wind	WSAvg	1.6	1.6	1.5	1.5	1.3	1.3	1.3	1.9	1.9	1.9	1.7	1.6	1.4
Precipitation	PrecAvg	3274	236	235	175	178	235	242	152	254	442	467	395	262
	PrecMax	5370	612	594	259	456	576	499	404	518	1364	1704	1020	604
	PrecMin	1794	66	40	53	38	56	67	6	68	68	62	58	53
	PrecStd	781	126	130	49	107	135	92	106	148	278	349	257	163
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	26.1	28.1	30.2	31.8	33.2	34.2	34.2	33.2	30.9	29.2	26.9
		MCWB	20.7	22.4	23.7	26.2	27.6	28.0	28.0	28.4	27.9	25.9	25.3	23.5
	2%	DB	23.1	24.4	26.5	29.0	30.5	32.3	33.6	33.3	32.1	29.3	27.8	24.8
		MCWB	19.7	21.2	22.9	25.1	26.7	27.7	28.0	28.2	27.2	25.3	23.9	21.5
	5%	DB	21.8	22.9	25.1	27.9	29.2	31.8	33.0	32.8	31.2	28.3	26.2	23.1
		MCWB	18.8	19.8	21.8	24.6	25.9	27.5	27.9	28.0	27.0	24.5	22.7	20.6
	10%	DB	20.1	21.1	23.8	26.6	28.2	30.9	32.2	32.0	30.2	27.0	24.8	22.0
		MCWB	17.9	18.8	21.1	23.7	25.2	27.1	27.6	27.6	26.5	23.8	22.0	19.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	23.0	24.3	26.7	28.1	29.3	29.9	29.6	29.1	27.1	25.9	23.9
		MCDB	23.8	25.8	27.2	29.4	30.9	32.2	32.7	32.7	32.1	29.5	28.6	26.0
	2%	WB	20.3	21.5	23.3	25.7	27.0	28.2	29.1	29.0	28.1	26.1	24.4	22.1
		MCDB	22.6	23.8	26.0	28.4	29.8	31.4	32.4	32.4	30.9	28.4	26.8	23.7
	5%	WB	19.2	20.2	22.3	24.8	26.2	27.7	28.4	28.5	27.5	25.4	23.4	21.0
		MCDB	21.1	22.4	24.6	27.5	28.9	30.9	31.9	31.8	30.3	27.2	25.4	22.5
	10%	WB	18.2	19.0	21.3	23.9	25.5	27.1	27.9	27.9	26.9	24.6	22.5	20.2
		MCDB	19.9	20.7	23.3	26.4	28.0	30.2	31.4	31.2	29.7	26.3	24.4	21.9
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.5	6.2	6.4	6.4	6.6	7.4	6.9	6.4	5.5	6.0	5.5
		MCDBR	8.3	8.7	8.6	8.0	7.8	7.7	8.0	7.8	7.7	7.7	8.0	8.1
		MCWBR	6.1	6.5	6.2	5.7	5.5	4.6	4.9	4.8	5.1	5.3	5.7	6.1
	5% WB	MCDBR	7.6	8.2	8.4	7.6	7.4	7.3	7.8	7.3	7.2	6.4	7.4	7.4
		MCWBR	5.9	6.6	6.3	5.9	5.4	4.8	5.4	5.0	5.2	5.1	5.8	6.0
Clear-Sky Solar Irradiance	taub	0.541	0.572	0.668	0.673	0.593	0.553	0.518	0.534	0.521	0.528	0.529	0.542	
	taud	1.898	1.864	1.674	1.715	1.935	2.050	2.156	2.101	2.120	2.039	2.008	1.918	
	Ebn at Noon	701	716	671	681	736	761	787	776	777	745	709	679	
	Edh at Noon	176	194	246	241	192	170	153	161	155	160	156	166	
All-Sky Solar Radiation	RadAvg	2.24	2.58	3.13	3.53	4.14	4.85	5.86	5.30	4.57	3.65	2.62	2.21	
	RadStd	0.40	0.48	0.46	0.54	0.71	0.57	0.34	0.47	0.47	0.44	0.31	0.37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	N/A	+81